security, performance monitoring, data integrity, concurrency control and data recovery

So, a database system is a combination of a database model, a DBMS and a database.

MySQL

MySQL is an open source, multi-threaded, relational database management system (RDBMS) written in C and C++. MySQL was created by a Swedish company, MySQL AB, founded by David Axmark, Allan Larsson and Micheal 'Monty' Widenius, and development started in 1994. The first version was launched on May 23, 1995. MySQL is named after co-founder Monty Widenius's daughter, My. The name of the MySQL dolphin logo, Sakila, was chosen from a huge list of names suggested by users in a 'Name the Dolphin' contest.

MySQL server is suitable for mission-critical, heavy-load production systems as well as for embedding into mass deployed software. It is regarded as a central component of the LAMP stack (Linux, Apache, MySQL and PHP). The MySQL database is available for various operating system platforms like Linux, Windows, MAC, BSD UNIX, etc. Popular websites like Wikipedia, Facebook and YouTube use MySQL.

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MySQL architecture

The MySQL server is based on a tiered architecture and consists of both primary sub-systems and support components that interact with each other to read, parse and execute queries to return the results.

The five types of primary sub-systems of MySQL are:

- Query engine
- Storage manager
- Buffer manager
- Transaction manager
- Recovery manager

Query engine: This contains three main interrelated components—the syntax parser, the query optimiser and the execution component. The syntax parser decomposes the SQL commands it receives from calling programs in a form that is understood by the MySQL engine. The 'query optimiser' streamlines the syntax for use by the execution component and then prepares the most efficient plan for query execution. The execution component interprets the execution plan, based on the information it receives and makes requests to other components to retrieve the records.

The storage manager: This interfaces with the operating system to write data to the disk in the form of user tables, indices and logs as well as internal system data.

The query cache: The MySQL engine uses an extremely efficient result set caching mechanism called query cache, which dramatically enhances the response time for queries that are called upon to retrieve the same data as the previous query.

The buffer manager: This handles all memory management issues between requests for data by the query engine and the storage manager. MySQL makes use of memory to cache result sets that can be returned, and the cache is maintained in the buffer manager.

The transaction manager: This sub-system provides a locking facility to ensure that multiple users access the data in a consistent manner without corrupting or damaging the data.

The recovery manager: This keeps copies of data for the purpose of retrieval in case of any sort of data loss.

The two support components of MySQL are:

- Process manager
- Function libraries

Process manager: This performs two main functions —managing user connections via modules for network connection management, and synchronisation of tasks and processes via modules for multi-threading, thread locking and performing thread-safe operations.

Function libraries: This contains general-purpose routines like string manipulation, sorting operations and performing OS-specific functions like memory management and file I/O.

Features of MySQL

- ***Relational database management system:*** MySQL supports all features, which makes it a complete RDBMS system. It supports full SQL as a standardised language for querying and updating data, and to administer the database.
- ***Easy and secure:*** MySQL is very simple and interactive to use, compared to other DBMS software, and is also highly secure with a solid data security layer providing efficient encryption to data.
- ***Client/server architecture:*** Its simple client/server architecture helps end users to create a server with connectivity to many clients, in order to communicate with the server for inserting, updating and administrating the database.
- ***Scalable:*** MySQL can handle high volumes of data without any hiccups—as much as 50 million rows. It can handle up to 8TB of data without any issues.
- ***Cross-platform:*** It is compatible with almost every operating system, like UNIX, Windows, Linux, MAC OS X, etc.
- ***High-performance, flexible and good productivity:*** MySQL provides faster, highly reliable, cheap storage solutions and supports a large number of embedded applications. It makes use of triggers, procedures and views to deliver better productivity.